



ME 207: Fluid Dynamics

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Lab Experiment 5: Working with flow meters

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Group 14

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Abstract

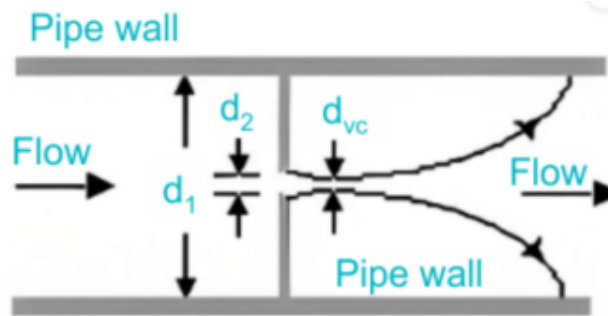
The aim of the experiment is to measure the flow rate of the fluid using different flow meters and comment on the accuracy of major devices used in the industry.

1 Task 5.1 (Essential Background)

Q1: Read up on the working principles of (i) an Orificemeter, (ii) a Venturimeter, (iii) a Nozzlemeter, and (iv) a Rotameter.

To find a flow rate through a pipe of the fluid, we constrict the flow and measure the pressure difference between two points using a differential pressure manometer. Flow meters, which are based on this principle, are called obstruction flow meters. The most commonly used obstruction flow meters are the orifice flow meters, nozzle flow meters and venturi flow meters.

Orifice meter:-

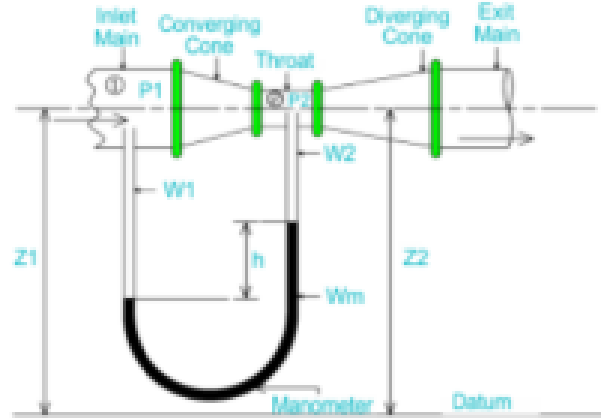


Working principle of an orifice meter :-

In the orifice flow meter, the fluid passes through a pipe that has a diameter of d_1 . The orifice plates are fixed in the pipe to obstruct the flow. The orifice plates are also called the restriction plates. These plates are holed and thin. Pressure gradually decreases when the fluid moves towards the orifice plate; after that, it increases. The pressure decreases till the vena contracta and, after that, it increases till the maximum. Vena contracta is the smallest area (diameter of vena contracta - d_{vc}) in the downstream of the orifice and where velocity is maximum. The orifice diameter is d_2 . The pressure increases, and the velocity decreases after the fluid passes through the vena contracta. This is the pressure difference, and the differential pressure measurement principle can do the measurement of the pressure difference.

The orifice flow meter follows the differential pressure measurement principle. The differential pressure measurement principle is that the differential pressure of the orifice meter is proportional to the flow rate of the fluid. The orifice flow meter follows Bernoulli's principle: The fluid's total energy per unit mass is constant. The sum of kinetic, potential and pressure energy remains constant. When the fluid's velocity increases, the fluid's pressure decreases, or the fluid's potential energy decreases. The height difference in the U-tube manometer measures the pressure difference.

(ii) Venturi meter:-



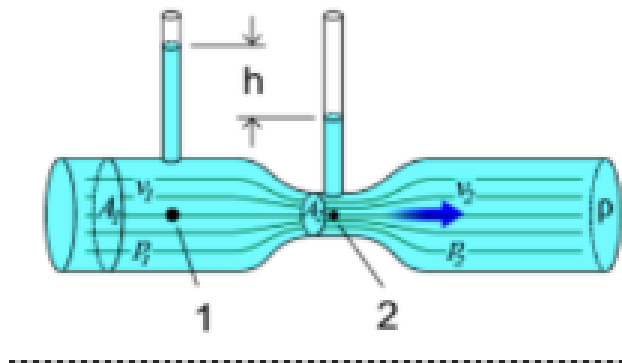
The venturi flow meter has three major parts or components: 1) converging part or cone, 2) Throat, and 3) Diverging part or cone.

1) Converging Part or cone:- The short portion of the pipe converges from the diameter of the pipe into the diameter of the throat. This short portion of the pipe is known as the converging part or cone.

2) Throat:- This is the pipe's small portion with small diameter and constant length. It is connected to the converging and diverging parts.

3) Diverging Part:- In the short portion of the pipe, the throat's diameter steadily increases up to the diameter of the pipe. This short portion is known as the diverging part or cone.

Working principle of Venturi meter :-

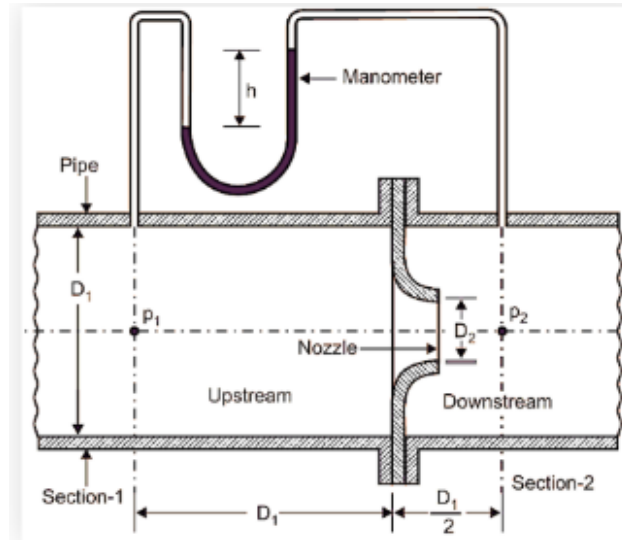


The fluid enters in the inlet area A_1 with velocity V_1 and pressure P_1 . If the pressure is high, then the flow goes into the capillary tubes. Capillary tubes are placed on the top of the venturi meter. These tubes are used to reduce or control the pressure. At the end of the converging part, velocity increases, and the pressure decreases. When the fluid passes through the smallest area (or the area of the throat- A_2), The velocity increases, and pressure decreases. This is called the venturi effect. At the end of the diverging part, the fluid's velocity decreases, and the pressure increases to its initial pressure.

The fluid in the capillary tube has a height difference (as shown in the above figure; section 1 has a height of h_1 in the capillary tube, and section 2 has a height of h_2 in the capillary tube), where section 1 is denoted in the inlet area of the venturi flow meter in the figure. Section 2 is denoted in the area of the throat in the figure, which is used to measure the pressure

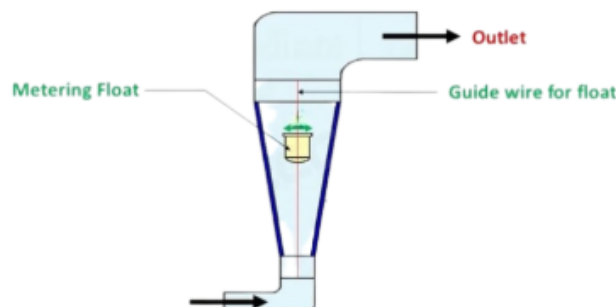
difference. We can also use a manometer to measure the difference in the pressure. The venturi flow meter follows the differential pressure measurement principle. The venturi flow meter follows Bernoulli's principle: The fluid's total energy per unit mass is constant. The sum of kinetic, potential and pressure energy remains constant. When the fluid's velocity increases, the fluid's pressure decreases, or the fluid's potential energy decreases.

(iii) Nozzle meter:-



Working principle of Nozzle meter:- The nozzle and the orifice flow meters are similar, but the nozzle flow meter uses a converging nozzle instead of the orifice plate with a hole. The fluid passes through the pipe, whose flow rate is to be measured. As shown in the figure, the shape of the nozzle is like a convergent inlet, and the end of the nozzle is connected to the throat. Due to the nozzle, The pressure drop occurs in the pipe. The nozzle flow meter follows the differential pressure measurement principle. To measure this pressure difference ($P_1 - P_2$), the U-tube manometer tap is attached about a D_1 diameter distance from the face of the nozzle along the upstream, and the other tap is attached about half of the diameter D_1 distance from the face of the nozzle along the downstream. This U-tube manometer gives us the height difference; through this, we can measure the pressure difference.

iv) Rotameter:-



Working principle of Rota meter:-

The rota flow meter consists of a hollow tapered glass tube. Another important part of the rota flow meter is the float. The float must be heavier than the fluid. The float density must be greater than the density of the fluid.

A fluid stream goes upward to the float from the bottom of the rota flow meter. The float rises due to the velocity head of the fluid and the buoyancy force of the fluid. The flow rate is higher if the float has a high tube height. The movement of the float is proportional to the flow rate of the fluid and the annular area between the float and the tapered tube. The annular area increases as the float moves up, and the pressure across the float decreases. The flow rate can be measured at a point where the float's weight balances by the fluid's velocity head and the fluid's buoyancy force.

Q2: What are the working principles of (i) an Electromagnetic Flow meter and (ii) an Ultrasonic flow meter?



(i) Working principle of an Electromagnetic Flow meter:-

The electromagnetic flow meter follows the principle of Faraday's magnetic induction. The principle states that changes in the magnetic field produce the electromotive force (emf) across an electrical conductor. The flow meter has a pair of electromagnets that generate the magnetic field. A conductive fluid passes through the field and generates a voltage at the right angles to the magnetic field and velocity vectors. To detect the resulting signal voltage, the electrodes are placed at the diameter of the pipe. The electrodes sense the voltage and are further processed by the transmitter, which gives a standardised output signal. When the profile is axisymmetric, the signal voltage is directly proportional to the average axial velocity.

The generated voltage is not affected by the fluid's physical properties like temperature, pressure, viscosity, density, etc.

(ii) Working principle of an Ultrasonic Flow meter:-



The ultrasonic flow meter measures the flow rate by observing the fluid flow on an ultrasonic beam. This meter uses sensors for upstream and downstream; pipes and reflectors are also used. This flow meter works on the principle that uses sound waves to find the velocity of the fluid in the pipe. The ultrasonic flow meters have two types: transit time and Doppler-effect (or frequency shift) flow meters. There are two flow, and no fluid flow cases into the pipe. The first case is where the ultrasonic wave is transmitted into the pipe, and the reflection from the fluid is similar. In the second case, the reflected wave's frequency is different due to the Doppler effect. The Doppler effect changes the frequency because the source and observer move towards (or away from) each other. For no-flow conditions, which means the fluid is not flowing, the time the signal takes to travel upstream is equal to the time the signal takes to travel downstream. In both conditions, the upstream wave will have a lower rate than the downstream because the wave moves opposite to the flow of the fluid. So, it takes much more time to reach than the wave, which moves with the flow. If the fluid moves faster, the difference between upstream and downstream time increases. The transmitter sends electrical signals to the transducers, which convert them into ultrasonic waves. The transmitter determines the flow rate using the upstream and the downstream times.

The transit time flow meter transmits sound waves in the upstream and the downstream directions and measures the difference in travel time. It features two transducers that take turns sending and receiving ultrasonic waves, one oriented in the flow direction and the other facing the opposite direction. The magnitude of average flow velocity V in the pipe is proportional to this travel time difference, t , and is determined from

$$V = K L t$$

Where:

- L is the distance between the transducers,
- K is the constant.

Q3: What is “Coefficient of discharge” (C_d), for an Orificemeter, a Venturimeter and a Nozzlemeter?

The derivation of discharge is the same for all three flow meters. Because all three use the differential pressure measurement principle to measure the pressure difference. However, the term A_2 is changing because the area of section 2 is different from one to another.

For an orificemeter:- A_2 is the area of the vena contracta,

For a venturi flow meter:- A_2 is the area of the throat,

For a nozzle flow meter:- A_2 is the area of the nozzle throat.

Coefficient of contraction:- The coefficient of contraction is the ratio of the area of the fluid near the vena contracta (A_{VC}) to the area of the (orifice/throat/nozzle throat). C_c denotes it.

$$C_c = \frac{A_{VC}}{A_o}$$

Coefficient of Velocity:- The coefficient of velocity (C_v) is the ratio of the fluid's actual velocity (V_{AC}) at the vena contracta to its theoretical velocity.

The theoretical velocity (V_{th}) is,

$$V_{th} = \sqrt{2gh}$$

Where,

- g = acceleration due to gravity,
- h = depth of the fluid

The coefficient of velocity is,

$$C_V = \frac{V_{AC}}{\sqrt{2gh}}$$

Coefficient of Discharge:- The coefficient of discharge (C_d) is the product of the coefficient of contraction (C_c) and velocity (C_v). It is also defined as the ratio of the actual discharge to the theoretical discharge.

$$C_d = C_V C_C$$

The derivation of the coefficient of discharge:-

To derive the coefficient of discharge,

Given that,

- P_1, P_2 : Pressure at sections 1 and 2,
- V_1, V_2 : Velocity at sections 1 and 2,
- Z_1, Z_2 : Height of fluid at sections 1 and 2,
- A_1 : Inlet Area of the pipe,
- ρ : Density of the fluid,
- g : Acceleration due to gravity.

Taking some assumption that the fluid is incompressible, ideal and inviscid.

We can apply Bernoulli's principle, which states that the sum of kinetic, potential and pressure energy remains constant.

$$\frac{P_1}{\rho g} + \frac{V_1^2}{2g} + Z_1 = \frac{P_2}{\rho g} + \frac{V_2^2}{2g} + Z_2$$

$$\left(\frac{P_1}{\rho g} + Z_1\right) - \left(\frac{P_2}{\rho g} + Z_2\right) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

The distance from the ground is same. ($Z_1 = Z_2 = 0$)

$$\frac{P_1}{\rho g} - \frac{P_2}{\rho g} = \frac{V_2^2}{2g} - \frac{V_1^2}{2g} \quad \dots(A)$$

The differential head (h_d) is,

$$h_d = \frac{P_1}{\rho g} - \frac{P_2}{\rho g}$$

From the equation-A,

$$(h_d) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

In the continuity equation, the mass flow rate is constant. For an incompressible fluid,

$$A_1 V_1 = A_2 V_2$$

$$V_1 = \frac{A_2 V_2}{A_1}$$

Substituting V_1 in the Bernoulli equation:-

$$(h_d) = \frac{V_2^2}{2g} - \frac{V_1^2}{2g}$$

$$(h_d) = \frac{V_2^2}{2g} - \frac{\left(\frac{A_2 V_2}{A_1}\right)^2}{2g}$$

$$(h_d) = \frac{V_2^2}{2g} \left(1 - \frac{A_2^2}{A_1^2}\right)$$

V_2 is,

$$V_2^2 = \frac{2gh_d}{\left(1 - \frac{A_2^2}{A_1^2}\right)}$$

$$V_2 = \sqrt{\frac{2gh_d}{\left(1 - \frac{A_2^2}{A_1^2}\right)}}$$

V_2 is the theoretical velocity.

In the real world, the flow is affected due to the friction, turbulence, etc. The coefficient of discharge is introduced to measure this loss.

$$C_d = \frac{C_C}{\sqrt{1 - \frac{A_2^2}{A_1^2}}}$$

The actual velocity at the vena contracta is,

$$V_{act} = C_d \sqrt{\frac{2gh_d}{(1 - \frac{A_2^2}{A_1^2})}}$$

The discharge rate (Q) is the product of the velocity at the vena contracta and the area of the vena contracta. To find the actual discharge, we should know the coefficient of discharge, which denotes the ratio of the actual discharge to the theoretical discharge.

$$Q = V_{Act}A_2$$

The formula of the discharge rate through an orifice meter is,

$$Q = C_d A_2 \sqrt{\frac{2gh_d}{(1 - \frac{A_2^2}{A_1^2})}}$$

Q4: Typically, the value of $C_d < 1$. What are the probable reasons for this?

The value of the coefficient of discharge (C_d) is < 1 . because it measures the real-world energy losses. The reasons are the following:-

1) In the orifice experiment, the fluid flows through the orifice, and it experiences friction from the wall of the pipe and the orifice.

Due to this friction, the velocity is decreased because the fluid's kinetic energy is converted to heat.

2) When the fluid passes through the orifice, the contraction and expansion form some eddies and turbulence. This thing also decreases the velocity of the fluid.

3) In the real world, the contraction is not ideal, so the coefficient of contraction measures this imperfection.

$$C_C = \frac{A_2}{A_O}$$

Where,

- A_2 = Area of the vena contracta,
- A_O = Area of the orifice

Since $A_2 < A_1$, $C_C < 1$.

4) The actual velocity of the fluid at vena contracta is less than the theoretical velocity due to energy losses. So, the coefficient of velocity measures this difference.

$$C_v = \frac{V_{act}}{V_2}$$

Where,

- V_{act} = actual velocity at the vena contracta,
- V_2 = theoretical velocity at the vena contracta.

Since $V_{act} < V_2$, $C_v < 1$

5) The coefficient of discharge (Cd) is the product of the coefficient of contraction (Cc) and velocity (Cv). So that,

$$C_d = C_c C_v$$

Since both $C_C < 1$ and $C_V < 1$, the product C_d is also less than 1.

2 Introduction

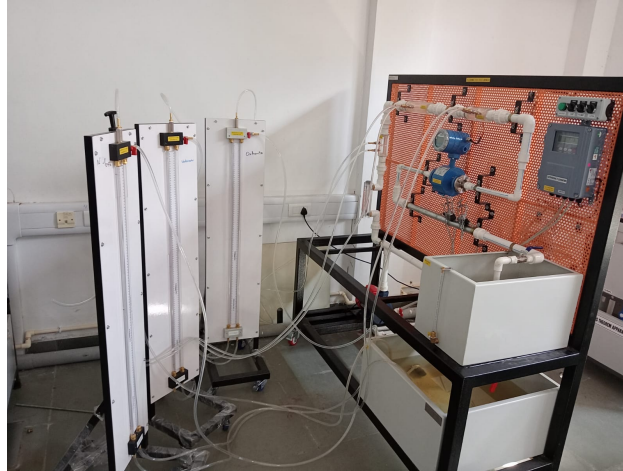


Figure 1: Experimental setup

For any fluid flowing through the pipe, the flow rate is a fundamental property that is directly related to the amount of fluid passing through the cross-sectional area of the pipe. Several devices, such as the orifice meter, nozzle meter, and venturi meter, are used to measure the flow rate. All these devices work on the differential pressure flow measurement principle, which states that the velocity of flow is directly related to pressure. This type of device is also known as an obstruction flow meter. On the other hand, there are other types of

flowmeters, such as electromagnetic flow meters and ultrasonic flow meters.

In this experiment, we measured the flow rate of fluid through these various meters and compared it with the actual flow rate calculated using elevation in the measuring tank. We see how the values of flow rates measured from the different flow meters indicated above deviate from the actual value and determine the accuracy of the above devices.

3 Objectives

- (i) Get hands-on experience with various types of flow meters used in industrial and laboratory applications.
- (ii) Learn about the working principles of various flow meters.
- (iii) Quantification of characteristic constants with analog flow meters (such as venturimeter, orificemeter, etc.)

4 Experimental Procedure

The following steps were taken to perform the experiment:

1. In this experiment, we measure the theoretical and experimental volumetric flow rates and define their relationship
2. Also, with the help of the Bernoulli equation, we'll measure the value of the difference in velocity of the fluid in the venturi flowmeter and nozzle flowmeter. Then, we'll compare this theoretical velocity to the experimental velocity.
3. There will be three values of experimental flow rate, which we'll fix in this experiment (0.7,0.8,0.9).
4. We'll adjust the valve to measure the value of the flow rate(experimental) of the fluid.
5. Then, according to these values, theoretical values are measured with the help of a tank. We'll close the valve below the tank to find the theoretical flow rate. And measure the time that fluid takes to reach a particular height in the tank.
6. Divide the time by that height and multiply this result by the tank area, which will give the experimental flow rate of the fluid. We'll take height as 3 cm to measure the value of the theoretical flow rate.
7. In Utube, the height difference will be measured to find the velocity difference between two points in the venturi flowmeter and nozzle flowmeter.
8. We'll repeat all these steps with three values of flow rate.

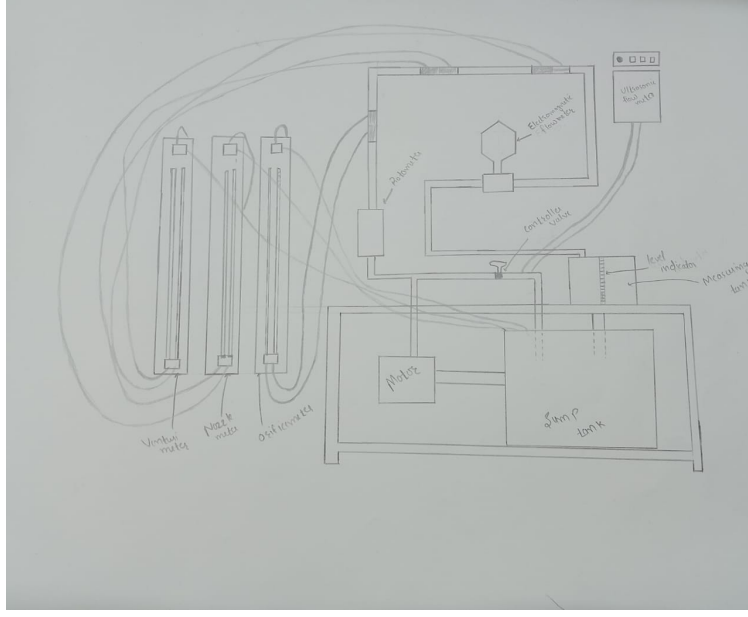


Figure 2: Schematic diagram of the experimental setup.

5 Results and Error Analysis

5.1 Theoretical flow rates

Before we measure the flow in the various flow-meters, we measure the actual flow rate through the pipe. The table below shows the exact flow rate through the pipe. To calculate the flow rate, we calculate the time taken for the sink tank to fill to a certain height. The

dimensions of the tank are the following.

- Length - 50 cm
- Breadth - 30 cm

Volume is measured in cm^3

Flow rate is measured in m^3/hr

Table 1: Flow Measurement Data - 1

Sr No.	Time(s)	Height(cm)	Volume	Flow Rate	Electric meter	Ultrasonic Measurement
1	22.01	3	4500	0.74	0.706	0.695
2	22.32	3	4500	0.73	0.706	0.695
3	22.76	3	4500	0.71	0.706	0.695

From the above table, we get the following reading.

- **Time average:** 22.36 s
- **Flow rate average:** $0.73 \text{ m}^3/\text{hr}$
- **Actual flow meter:** $0.706 \text{ m}^3/\text{hr}$
- **Percentage error:** 3.4 %

For the actual flow meter reading, we are considering the values from the electric meter reading because the ultrasonic meter takes a lot of time to calibrate, and while controlling the valve, we pay attention to the electric meter to adjust the flow rate so that is one more reason why we include those values. We repeat the experiment after 30-40 minutes to check whether the flow rate has changed, which should not change because it does not depend on time. The same experiment was repeated, and the values we obtained have been tabulated below.

Table 2: Flow Measurement Data - 2

Sr No.	Time(s)	Height(cm)	Volume	Flow Rate	Electric meter	Ultrasonic Measurement
1	24.27	3	4500	0.67	0.69	0.61
2	21.82	3	4500	0.77	0.69	0.61
3	24.06	3	4500	0.67	0.69	0.61

From the above table, we get the following reading.

- **Time average:** 23.38 s
- **Flow rate average:** $0.7 \text{ m}^3/\text{hr}$
- **Actual flow meter:** $0.69 \text{ m}^3/\text{hr}$
- **Percentage error:** 1.45 %

From the above two experiments, the mean flow rates for the first and second experiments are equal to be equal to **$0.715 \text{ m}^3/\text{hr}$** and the standard deviation is equal to **0.02**. As they are very close to each other, we can say that the flow rate does not vary with time.

Table 3: Flow Rate Averages for flow rate $0.7 \text{ m}^3/\text{hr}$

Experiment Number	Flow rate average
1	0.73 ± 0.02
2	0.7 ± 0.02

We now repeat the same procedure two times for flow rates $0.8 \text{ m}^3/\text{hr}$ and $0.9 \text{ m}^3/\text{hr}$.

Table 4: Flow Measurement Data - 1 for flow rate $0.8 \text{ m}^3/\text{hr}$

Sr No.	Time(s)	Height(cm)	Volume	Flow Rate	Electric meter	Ultrasonic Measurement
1	21.14	3	4500	0.77	0.8	0.78
2	19.81	3	4500	0.82	0.8	0.78
3	20.31	3	4500	0.8	0.8	0.78

From the above table, we get the following reading.

- **Time average:** 20.42 s
- **Flow rate average:** $0.797 \text{ m}^3/\text{hr}$
- **Actual flow meter:** $0.8 \text{ m}^3/\text{hr}$
- **Percentage error:** 0.38 %

The same experiment was repeated, and the values we obtained have been tabulated below

Table 5: Flow Measurement Data - 2 for flow rate $0.8 \text{ m}^3/\text{hr}$

Sr No.	Time(s)	Height(cm)	Volume	Flow Rate	Electric meter	Ultrasonic Measurement
1	20.53	3	4500	0.79	0.8	0.77
2	19.32	3	4500	0.84	0.8	0.77
3	21.06	3	4500	0.77	0.8	0.77

From the above table, we get the following reading.

- **Time average:** 20.3 s
- **Flow rate average:** $0.797 \text{ m}^3/\text{hr}$
- **Actual flow meter:** $0.8 \text{ m}^3/\text{hr}$
- **Percentage error:** 0.38 %

From the above two experiments, the mean flow rates for the first and second experiments are equal to be equal to **$0.797 \text{ m}^3/\text{hr}$** and the standard deviation is equal to **0**. As they are equal, we can say that the flow rate does not vary with time.

Table 6: Flow Rate Averages for flow rate $0.8 \text{ m}^3/\text{hr}$

Experiment Number	Flow rate average
1	0.797 ± 0
2	0.797 ± 0

Table 7: Flow Measurement Data - 1 for flow rate $0.9 \text{ m}^3/\text{hr}$

Sr No.	Time(s)	Height(cm)	Volume	Flow Rate	Electric meter	Ultrasonic Measurement
1	19	3	4500	0.85	0.91	0.926
2	18.36	3	4500	0.88	0.91	0.926
3	17.65	3	4500	0.92	0.91	0.926

From the above table, we get the following reading.

- **Time average:** 18.34 s
- **Flow rate average:** $0.883 \text{ m}^3/\text{hr}$
- **Actual flow meter:** $0.91 \text{ m}^3/\text{hr}$
- **Percentage error:** 2.97 %

The same experiment was repeated, and the values we obtained have been tabulated below

Table 8: Flow Measurement Data - 2 for flow rate $0.9 \text{ m}^3/\text{hr}$

Sr No.	Time(s)	Height(cm)	Volume	Flow Rate	Electric meter	Ultrasonic Measurement
1	18.02	3	4500	0.9	0.89	0.95
2	18.71	3	4500	0.87	0.89	0.95
3	20.03	3	4500	0.81	0.89	0.95

From the above table, we get the following reading.

- **Time average:** 18.92 s
- **Flow rate average:** $0.86 \text{ m}^3/\text{hr}$
- **Actual flow meter:** $0.95 \text{ m}^3/\text{hr}$
- **Percentage error:** 9.47 %

From the above two experiments, the mean flow rates for the first and second experiments are equal to be equal to **$0.872 \text{ m}^3/\text{hr}$** and the standard deviation is equal to **0.02**. As they are very close to each other, we can say that the flow rate does not vary with time.

Table 9: Flow Rate Averages for flow rate $0.9 \text{ m}^3/\text{hr}$

Experiment Number	Flow rate average
1	0.883 ± 0.02
2	0.86 ± 0.02

From the above readings for all the six experiments we can confirm that theoretical flow rates are approximately the same as in the electromagnetic or the ultrasonic flow meter.

5.2 flow rates in Analog flow meters

The following derivation is applicable for venturimeter, manometer, and orificemeter.

$$P_1 + \rho gh + \frac{1}{2}\rho v_1^2 = P_2 + \rho gh_2 + \frac{1}{2}\rho v_2^2 \quad (1)$$

$$h_1 \approx h_2 \text{ in pipe.}$$

$$P_1 + \frac{1}{2}\rho v_1^2 = \frac{1}{2}\rho v_2^2 + P_2$$

$$P_1 - P_2 = \frac{1}{2}\rho(v_2^2 - v_1^2) \quad (2)$$

$$\rho g \Delta h = \frac{1}{2}\rho(v_2^2 - v_1^2)$$

$$2g\Delta h = v_2^2 - v_1^2$$

$$2g\Delta h = v_1^2 \left(\left(\frac{A_1}{A_2} \right)^2 - 1 \right)$$

$$2g\Delta h = v_1^2 \left(\left(\frac{d_1}{d_2} \right)^4 - 1 \right) \quad (3)$$

$$A_1 v_1 = \frac{\sqrt{2g\Delta h}}{\sqrt{\left(\frac{d_1}{d_2} \right)^4 - 1}} \times A_1 = A_2 v_2$$

The only assumption in the above equation is that the fluid is incompressible and the flow is steady, from which we get the formula for flow rate.

$$Q = \frac{\sqrt{2g\Delta h}}{\sqrt{\left(\frac{d_1}{d_2} \right)^4 - 1}} \times A_1 = A_2 v_2 \quad (4)$$

5.2.1 coefficient of discharge for a nozzle meter

Using the above formula we can calculate the actual flow rate through the manometer.

- $d_1 = 26 \text{ mm}$
- $d_2 = 13 \text{ mm}$
- $g = 9.81 \text{ m/s}^2$
- $A_1 = 0.00053066 \text{ m}^2$
- **water-tank flow rate = C_d * theoretical flow rate**

Table 10: Flow Measurement Data for nozzle meter with flow rate $0.715\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	86.8	97.9	11.1	0.73	0.715	0.98
2	82.7	93.5	10.8	0.72	0.715	0.99

Table 11: Flow Measurement Data for nozzle meter with flow rate $0.797\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	49.7	63.8	14.1	0.82	0.797	0.97
2	71.4	85.4	14	0.82	0.8	0.97

Table 12: Flow Measurement Data for nozzle meter with flow rate $0.872\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	49.2	67.3	18.1	0.93	0.872	0.94
2	37.7	55.9	18.2	0.93	0.872	0.94

- $h_1, h_2, \Delta h$ are in centimeters
- Actual flow rate, electric meter reading are in m^3/hr
- Coefficient of discharge is a unit less quantity.
- The mean value of coefficient of discharge is **0.965 ± 0.02**

5.2.2 coefficient of discharge for a venturimeter

Using the above formula we can calculate the actual flow rate through the manometer.

- $d_1 = 26\text{ mm}$
- $d_2 = 13\text{ mm}$
- $g = 9.81\text{ m/s}^2$
- $A_1 = 0.00053066\text{ m}^2$
- **water-tank flow rate = C_d * theoretical flow rate**

Table 13: Flow Measurement Data for venturimeter with flow rate $0.715\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	60	54.4	5.6	0.52	0.715	1.38
2	60	54.2	5.8	0.53	0.715	1.35

Table 14: Flow Measurement Data for venturimeter with flow rate $0.797\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	61.9	54.8	7.1	0.58	0.797	1.37
2	62	54.7	7.3	0.59	0.797	1.35

Table 15: Flow Measurement Data for venturimeter with flow rate $0.9\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	64.5	55.3	9.2	0.66	0.872	1.32
2	64.3	55	9.3	0.67	0.872	1.3

- $h_1, h_2, \Delta h$ are in centimeters
- Theoretical Flow Rate and water-tank flow rate reading are in m^3/hr
- Coefficient of discharge is a unit less quantity.
- The mean value of coefficient of discharge is 1.345 ± 0.03

5.2.3 coefficient of discharge for a orifice meter

Using the above formula we can calculate the actual flow rate through the manometer.

- $d_1 = 26\text{ mm}$
- $d_2 = 13\text{ mm}$
- $g = 9.81\text{ m/s}^2$
- $A_1 = 0.00053066\text{ m}^2$
- **water-tank flow rate = C_d * theoretical flow rate**

The table below gives the value for coefficient of discharge for flow rate of $0.7\text{m}^3/\text{hr}$

Table 16: Flow Measurement Data for orifice meter with flow rate $0.715\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	77.8	51.3	26.6	1.13	0.714	0.63
2	61.6	35.4	26.2	1.12	0.715	0.64

Table 17: Flow Measurement Data for orifice meter with flow rate $0.797\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	88.7	55.1	33.6	1.27	0.797	0.63
2	63.1	28.4	34.7	1.29	0.797	0.62

Table 18: Flow Measurement Data for orifice meter with flow rate $0.9\text{m}^3/\text{hr}$

Reading No.	h_1	h_2	Δh	Theoretical Flow Rate	water-tank flow rate	C_d
1	61.5	17.6	43.9	1.45	0.872	0.6
2	77.6	34.5	43.1	1.43	0.872	0.61

- $h_1, h_2, \Delta h$ are in centimeters
- Theoretical Flow Rate and water-tank flow rate reading are in m^3/hr
- Coefficient of discharge is a unit less quantity.
- The mean value of coefficient of discharge is **0.62 ± 0.015**

5.2.4 coefficient of discharge for a rotameter

A rotameter has a piston inside it which is used to calculate the readings of flow rate by measuring the point coinciding with the top surface of the piston.

The least count of a rotameter is very high so the coefficient of discharge and the flow rate will not be accurate.

Table 19: Rotameter Vol. Flow Rate Data - 1

Sr No.	Vol. flow rate(m^3/hour)	Actual reading (m^3/hour)	Percentage error
1	0.88	0.715	23.07692308
2	0.9	0.797	12.92346299
3	1.1	0.972	13.16872428

We repeated the experiment and got the following data

Table 20: Rotameter Vol. Flow Rate Data - 2

Sr No.	Vol. flow rate(m ³ /hour)	Actual reading (m ³ /hour)	Percentage error
1	0.88	0.715	23.07692308
2	1	0.797	25.47051443
3	1.1	0.972	13.16872428

Now we calculate the coefficient of discharge for flow rate 0.715(m³/hour).

Table 21: Coefficient of Discharge for three different flow rates

Vol. flow rate (m ³ /hour)	Water tank reading(m ³ /hour)	Coefficient of discharge
0.88	0.715	0.81
0.88	0.715	0.81
0.9	0.797	0.8
0.9	0.797	0.89
1.1	0.872	0.79
1.1	0.872	0.79

The mean value of coefficient of discharge is **0.815 ± 0.04**

5.3 Causes of error

1. **Viscous losses** - As fluid flows through the pipe, some Kinetic energy is lost because of the interaction between fluid and the pipe walls. These losses are accounted into viscous losses.
2. **Losses due to valves and bends** - Components like joints and valves can cause turbulence in the flow and lead to a loss of kinetic energy; such losses are hard to determine and cause a mismatch in the results.
3. **Response error** - When we measured the time for the specific height reached in the tank to determine the flow rate, the time was measured manually with the help of a watch. So, it can lead to a delay in time measurement.
4. **Volumetric flow rate is not constant** - In the experiment, the flow rate is not constant because the flow rate is fluctuating in the electromagnetic flow meter. We experimented based on the assumption that the flow rate is constant throughout the experiment.
5. **Instrumental inaccuracies** - There were minor defects in the experimental apparatus, such as the measuring scale being bent, which can lead to inaccurate readings and cause errors.

6. **Imperfections in fluid** - The fluid used to experiment was water. However, some impurities, such as dirt and air bubbles in the fluid, could contribute to the total error.
7. **Least count errors** - The number called least count is associated with every measuring instrument; errors arising from the least count are termed as least count errors. The contribution of the least count errors to the total error depends upon the instrument's resolution.
8. **Parallax error** - This error is caused when the observer takes measurements from an angle, resulting in differences from the actual reading.
9. **Rotameter least count** - The least count of a rotameter is very high (600 Liter per minute); therefore, when the top surface of the cylinder inside the rotameter is in between the readings, it causes a lot of deviations from actual readings.
10. **Resolution of Camera** - While taking the readings, we clicked photos of the manometer tubes, as the camera's resolution wasn't good; a significant difference of 0.1 to 0.2 cm can be expected in the readings.
11. **Venturimeter coefficient of discharge** - The value of the venturi meter coefficient of discharge was greater than 1, which is wrong; the reason could be inaccurate height measurement readings, and the second reason is that the flow rate varies a bit during the experiment, but we consider it to be constant, which is incorrect.
12. **Considering constant height** - In the tube above the rotameter, we do consider the height to be steady, but there is an increase in the height of approx 2 cm, which we ignore, hence giving us an inaccurate measurement of the coefficient of discharge.

6 Observation

After measuring the flow rates for various devices and calculating the coefficient of discharge for each one, it was observed that the value is more significant than one for the venturi meter. At the same time, it was slightly less than one for the nozzle meter and significantly less than one for the orifice meter.

7 Conclusion

The unexpected value of the coefficient of discharge in the case of a venturi meter may arise because of uncertainties in measuring the value of flow rates, which are mentioned in the sixth section. It can be concluded that the kinetic losses are least in the Venturi meter, followed by the nozzle meter and orifice meter.

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